

M002 Extension 2 — Piston & Logic Maze

Topic: Two Pistons + AND Conditions · **Course:** Maze Madness · **Level:** Extension (Intermediate) ·

Time: about 45 minutes

The agent follows a **redstone trail** through a long maze. At some junctions it checks **two things at once** — an **AND** condition. Twice, a **lever powers a piston** that opens the way. Your job: get the agent through the **whole maze**.

You drive it with chat words: **l** turn left, **r** turn right, **run** start the solver, **r1** teleport back to you.

1 · Read the Chat Commands

Each block binds a **chat word** to an action.

```
def on_chat():
    agent.turn(RIGHT_TURN)
    player.on_chat("r", on_chat)
```

You type **r**. What does the agent do?


```
def on_chat():
    agent.teleport_to_player()
    player.on_chat("r1", on_chat)
```

You type **r1**. Where does the agent go? When does that help in a long maze?

2 · AND at a Junction

AND is true only when **both** parts are true.

```
if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, AHEAD):  
    agent.turn(LEFT_TURN)
```

Fill the table. Write turn or no turn.

Redstone LEFT?	Redstone AHEAD?	What happens?
yes	no	
no	yes	
yes	yes	

In your own words: what must be true for the agent to turn left?

3 · Trace the Solver 🔍

This loop runs when you type `run`.

```
while True:
    agent.move(FORWARD, 1)
    if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, AHEAD):
        agent.turn(LEFT_TURN)
    elif agent.detect(BLOCK, AHEAD):
        agent.interact(AHEAD)
        agent.move(FORWARD, 1)
```

The last `elif` is a piston gate. A block is ahead. What does `agent.interact(AHEAD)` do?

Why can the agent move forward right after it interacts?

4 · Fix the Bug 🐛

The agent should turn left only when redstone is left AND ahead. This turns on either one.

```
if agent.detect(REDSTONE, LEFT) or agent.detect(REDSTONE, AHEAD):  
    agent.turn(LEFT_TURN)
```

Hint: "left and ahead" — both at once.

Write the fixed line:

Why was it wrong?

run

at the

e age

de the

6 · Record Your Walkthrough 🎥

Film on a phone while the agent solves the maze. Talk like you are teaching a friend. Try to use: **redstone, trail, AND, piston, lever, loop.**

1. Show the maze and the redstone trail.
2. Type **run** and show the agent following it.
3. Read one AND check out loud and say what it decides.
4. Show the agent flipping a lever to open a piston.
5. Show it reaching the goal.

Write what you will say. Plan it here first.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.